



Follow-up and extension

Introduction to Remote Sensing

In this lesson, students will be introduced to Remote Sensing while using an MFworks exercise on a Landsat 5 image. They will investigate different spectral bands on a raster image. The following information can help you better understand what is Remote Sensing? This information is taken from the “Earth Observation Curriculum for High School, Geomatics for Educators”, made by the Image Centre and from the tutorial “Fundamentals of Remote Sensing” at the Canada Centre for Remote Sensing Website <http://www.ccrs.nrcan.gc.ca/ccrs/eduref/tutorial/tutore.html>. We also suggest that you use the PowerPoint presentation available in the Teacher Tools section of the drop down menu of this web site. This PowerPoint presentation was made in collaboration with the Image Centre and Claude Brun del Re (Woodroffe High School Teacher).

Definition of Remote Sensing

Remote Sensing is often defined as indirect (remote) observations (sensing), which yield information about a feature or **target**. The observations are made by **sensors**, which are measuring energy **reflected** or **emitted** from the target.

Common examples of sensors include our eyes and cameras. Our eyes sense and measure light – the energy of **visible wavelengths**. Other sensors can measure non-visible wavelengths. With knowledge and experience, interpreting our observations can lead to new information about the earth. Much more can be discovered about our surroundings and environment by examining the information contained within these other wavelengths of energy.

Some remotely sensed information includes ultrasounds, satellite weather maps, speed radar, graduation photos, and sonar (for ships and for bats!). Hospital use imaging technology, including CAT scans, magnetic resonance imaging (3-D imaging of soft tissue), x-rays – these are all non-intrusive, remote methods for examining our bodies.

What other examples of remote sensing can you think of? There is quite a lot that we use in our everyday life.

How does Remote Sensing work?

The sensing instruments are usually far removed from the objects being observed. The sensor resides on a **platform**, which could be a satellite, shuttle, aircraft, balloon, or ground based stage. The distance between the objects being recorded and the platform plays a large role in determining the detail of information obtained and amount of surface area imaged by the sensor.

Sensors onboard platforms far away from their targets typically view a larger surface area, but cannot provide great detail of that surface. Compare what an astronaut onboard the space shuttle sees of the earth to what you can see from an airplane. The astronaut might see your whole province or country in one glance, but couldn't distinguish individual houses. Flying over a city or town, you would be able to see individual buildings and cars, but you would be viewing a much smaller area than the astronaut would. There is a similar difference (though not as extreme) between satellites images and air photos.

The detail discernible in an image is called “**special resolution**”. **Special resolution** refers to the size of the smallest possible distinguishable feature on an image. Images where only large features are apparent are said to have **coarse resolution**. In **fine resolution** images, very small objects can be discerned. Military sensors used for espionage or reconnaissance activities are designed to vie as much detail as possible, and therefore have very fine resolution. They can

see a driving license plate. If you want to explore the Terraserver site at <http://www.terraserver.com>, you may see some fine resolution satellite images. This site was historically made of Russian spying images. For viewing very fine resolution satellite images explore the new IKONOS high-resolution satellite at <http://www.spaceimaging.com/>. In their gallery section you will see several beautiful images taken by this satellite and among them one at the ESRI convention where we see a huge ESRI sign with students taken from this satellite and the images of the World Trade Centre and the Pentagon before and after September 11, 2001.

Scale Differences

Commercial satellites provide imagery with resolutions varying from 1m to 5km. Finer resolution images usually cover less total ground area. The ratio of distance on an image or map, to actual ground distance is referred to as **scale**.

The following table demonstrate how platforms, resolution, and scale are related:

Platform	Scale	Ground Area Covered	Resolution
Satellite	small	large	medium to coarse
Aircraft	medium to large	small	fine
Ground-based	very large	very large	very fine

Targets

One of the primary goals of Remote Sensing is to examine targets to identify them and/or provide meaningful information about them. A target is any feature or object you wish to observe on an image.

The target

- may be a point, line, or area feature. This means it can have any form, from a large expanse of water, to a bus in a parking lot.
- is detectable. The target must contrast with other features around it, and therefore be distinguishable from its surroundings.
- must be illuminated by an energy source, or emit its own radiation.

Targets are defined in terms of their response to incident energy or their emission of energy (radiation). The response to incident radiation, in the form of reflected energy, is measured by a sensor, and is depicted in an image product. Air photos and satellite images are common examples of these products.

Examples of targets:

- Point: houses, industrial buildings
- Line: major roads, railroad, dam, bridges, airport
- Area: river, agricultural fields, forest

Observation versus Inference

Remember that a remotely sensed image provides data, not information. Information comes from the interpretation of the data, or the inferences made about that data. This is the basic difference between observations and inference – an observation is a simple comment or description about the data, the inference is the information provided by the observation. The inference comes from the interpreter's experience and knowledge of the sensor, the application, or the features on the ground and their interaction with the energy.

Interpretation Elements

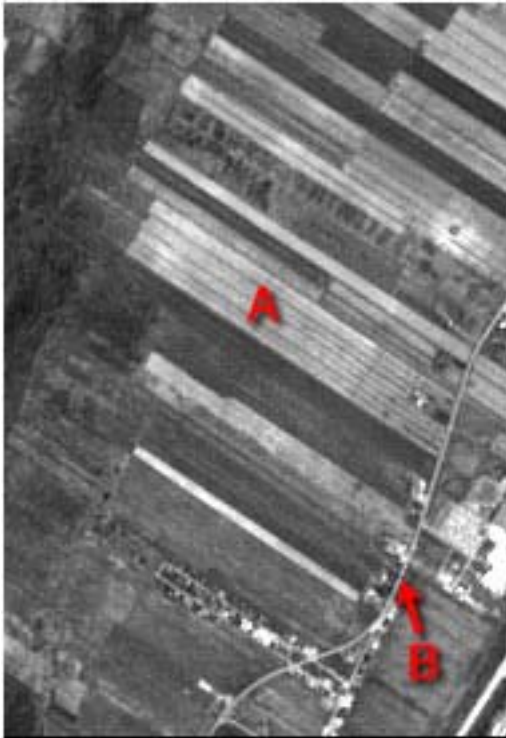
Recognizing targets is the key to interpretation and information extraction. Observing the differences between targets and their backgrounds involves comparing the differences in target **size, shape, pattern, tone, texture, and association** with other features. These parameters are referred as **interpretation elements**. Keep these elements in mind when comparing targets – see how they differ.

Most of the images below are taken from the Images of Canada on the web site of Canada Centre for Remote Sensing at:

http://www.ccrs.nrcan.gc.ca/ccrs/imgserv/tour/05/05_quee.html

There are mainly sub-scenes from a SPOT image of Montréal Quebec. The other image is a section from an air photo of Renfrew, Ontario.

Tone



Tone refers to the colour or relative brightness of objects. Generally, tone is the most important element for distinguishing features in imagery.

The image shown here is a small portion of an agricultural area outside the urban expanse of Montréal. It is simple to identify the early French Canadian method of subdividing agricultural lands. Lengthwise subdivision, eventually resulting in long narrow plots (A), allowed for efficient working of the fields (fewer turns) and for more access to lakes and rivers for transportation. While it is not possible to identify crop type in this image, it is possible to get an estimation of the area under cultivation, field sizes and field boundaries. The darker portions of land would be under some type of crop, or in pasture. The lighter areas represent little or no vegetation at the time the imagery was acquired (May 17). The smooth curve of the road (B) provides a good contrast to the razor straight edges of the field. Notice the houses scattered along the road.

Texture



Texture is the “smoothness” and consistency of tone. Smooth textures are the result of uniform, even surfaces, such as fields, asphalt, or grasslands. A target with an uneven surface and irregular structure such as a forest creates a rough appearance.

Shape



Shape refers to the general form, configuration, or outline of individual objects. Shape can be a very distinctive clue for interpretation. Straight edge shapes typically represent urban or agricultural (field) targets, while natural features, such as forest edges, are generally irregular, except where man has created road or clear cuts.

Several runways (**A**) of the St.-Hubert airport are clearly visible in this image. While not legible, some of the markings on the runways are actually visible! The smaller runways used to taxi planes are seen running parallel to the major runways. The very bright features nestled along side of the runways would be buildings associated with the airport (**B**). The thin grey line running around the exterior of the airport (**C**) is likely a perimeter access road, while (**D**) is a major highway. The St.-Hubert airport is adjacent to the site of the Canadian Space Agency Headquarters, as well as to RADARSAT mission control.

Pattern



Pattern is the spatial arrangement of visibly discernible objects. Typically a repetition of similar objects will produce a distinctive and ultimately recognizable pattern. Orchards and urban streets are examples of regular patterns.

The incredible resolutions that are available from some satellite imagery are evident in this small portion of the 10 metre SPOT data. A portion of the Trans Canada Highway (**A**) is visible in the south portion of the image. Individual cars and trucks can be identified on this road by their very bright reflections. A residential neighbourhood of Boucherville (**B**) is seen above the highway. Can you identify the major feeder roads that run through the suburb? These are the larger roads that feed traffic onto the individual residential streets. New roads and houses are under construction at (**C**) where the suburb is expanding. Below the highway is an industrial area with large, low buildings (**D**).

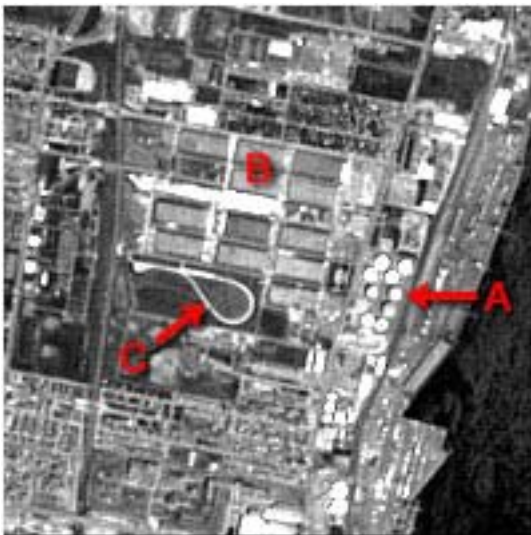
Size



Size is a function of scale. It is important to assess the size of a target relative to other objects in a scene, as well as an absolute size, to aid in the interpretation of that target. A quick approximation of target size can direct interpretation to an appropriate result more quickly. For example, if an interpreter had to distinguish zones of land use, and saw an area with a number of buildings in it, large buildings such as factories or warehouses would suggest commercial property, whereas small buildings would indicate residential use.

Montréal hosted the Summer Olympic Games in 1976 and today the facilities are used for both professional and recreational purposes. The site, primarily made of cement, is the extremely bright feature in the image. The Olympic Stadium can be identified by a slightly darker ring surrounding a white oval (**A**). The dark irregularity at the top of the oval is the shadow cast from the stadium's tower. The bright buildings and grounds around (**B**) are Montréal's Botanical Gardens, while the dark area above it (**C**) is Maisonneuve Park. Walking paths can be seen within the park.

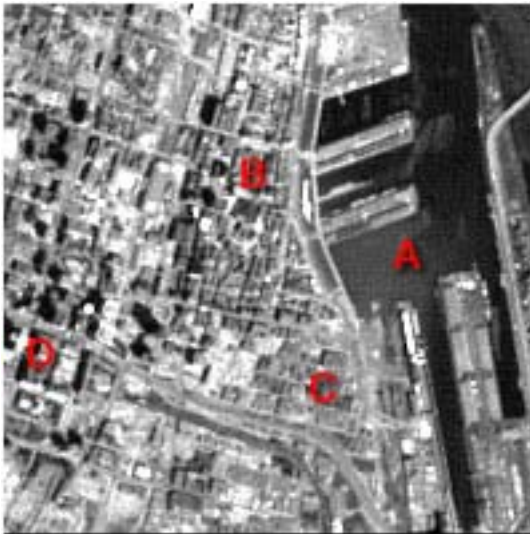
Association



The association of the other objects in proximity to the target of interest may provide information to facilitate interpretation.

An industrial part of Montréal is seen in this image. Along with the larger blocks and buildings (**B**) associated with an industrial area of town are large round tanks (**A**). The tank contents are likely to be oil, as there are several oil refineries in this section of the city. Notice the railway loop at **C**.

Shadow



Shadows present in imagery can aid greatly in the interpretation of shape and relative heights of targets. This can be very important in urban areas and for tree identification.

On the eastern side of this image, a major port (**A**) is identifiable. Two quays are visible in the east-west direction and one in the north south. The slightly darker section around **B** is one of the oldest districts in the city, complete with cobble stone streets. The lighter grey section (**C**) by the port is an industrial district with warehouses and factories. Some very large buildings are visible on the left side of the image (**D**), which happens to be the financial district of Montréal. The tall height of these buildings can be inferred from shadows that are being cast.

Interpretation Keys

An interpretation key is either a chart or table, which gives a general description of the characteristics of the targets to be identified, using the elements illustrated above. A key is essentially a guide for the interpreter, and facilitates interpretation accuracy and consistency. This is especially important when multiple interpreters, multiple photos, and multiple times of interpretation are used.

Here is an example of an interpretation key.

Target	Tone	Texture	Shape	Pattern	Size	Association	Shadow
Major roads							
Water bodies							
Residential areas							
Commercial/ industrial areas							
Recreational areas							

Energy Structure and Movement

Electromagnetic radiation behaves in predictable ways. Understanding the structure, movement and interaction of this type of energy facilitates the acquisition, processing and interpretation of remotely sensed imagery.

One of the most important characteristics of radiation to remember is that it has wave-like properties, and some of the basic properties of energy vary with **wavelength**. Knowing the wavelengths of various types of energy helps us to understand how the energy will interact with the atmosphere and the targets on the earth's surface.

When energy moves as a wave, two characteristics are particularly important for understanding the information inherent in remotely sensed imagery. These are the wavelength and frequency.

High sounds have short wavelengths and high frequencies. Low sounds are the opposite. Scientist says that the earth itself vibrates at a very low frequency, making a sound far below the humans' hearing range. Dogs can hear sounds of much higher frequencies than our ears can detect. This is analogous to being able to see past the visible range of human eyesight, into the ultraviolet portion of the spectrum.

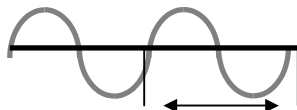
Sensors are designed to detect and measure very specific ranges of wavelengths. It is important to know the wavelength of the energy you want to study before deciding which sensor to use.

Light energy travels in waves



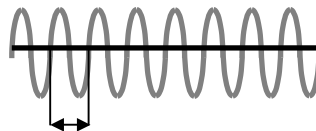
Wavelength: distance between crests of waves

Wavelength is a function of energy - longer wavelengths have less energy



wavelength

longer wavelength: less energy

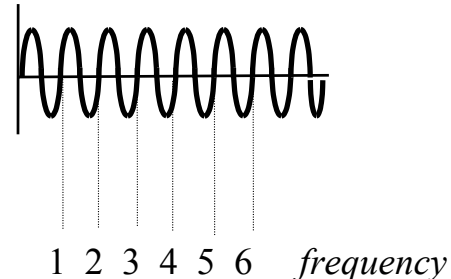
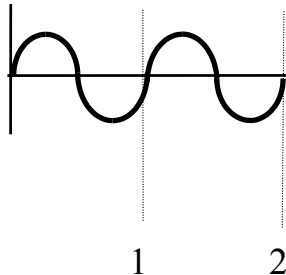


wavelength

shorter wavelength: higher energy

Frequency: number of cycles per unit of time

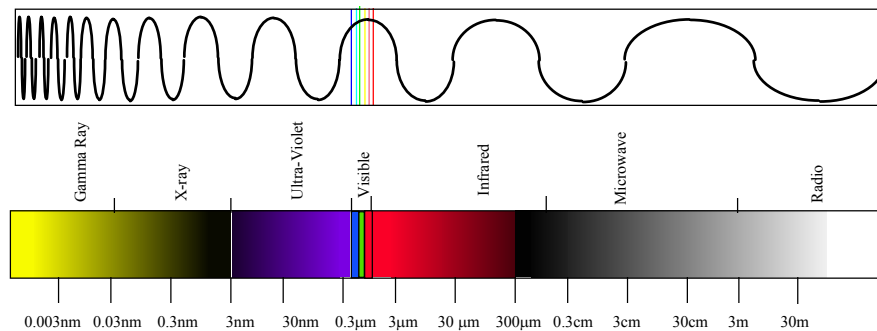
Frequency is inversely proportionally to wavelength - shorter wavelength : higher frequency



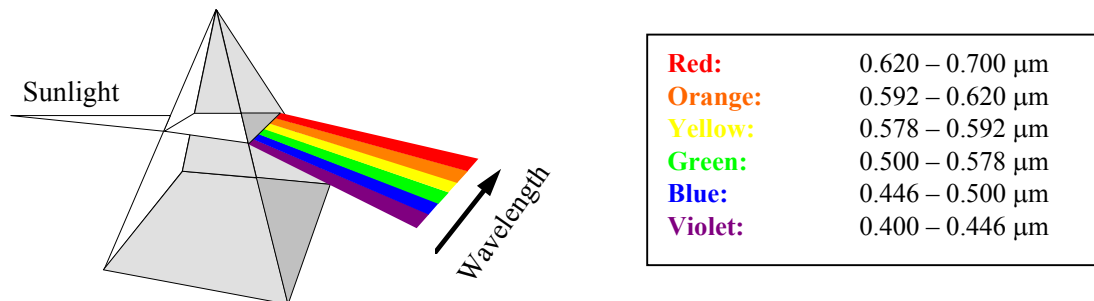
Electromagnetic Spectrum

The electromagnetic spectrum is a means of organizing and displaying types of electromagnetic radiation by wavelength. The shorter wavelengths (gamma and x-rays) are on the left of the chart, with longer microwaves and broadcast radio waves on the right.

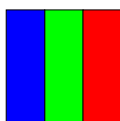
It is important to recognize how small the visible portion is relative to the rest of the spectrum. There is a lot of radiation around us which is “invisible” to our eyes, but which other sensors can detect.



The light that our eyes can detect is only a small portion of the electromagnetic energy that surrounds us. The wavelength range of visible radiation is approximately 0.4 to 0.7μm. This corresponds to the wavelengths of blue, green, and red light.



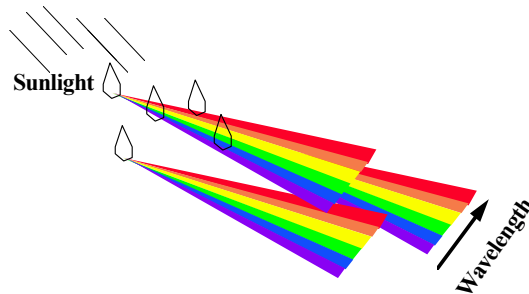
Although we see sunlight as a uniform or homogenous colour, it is actually composed of various wavelengths of energy in primarily the infrared, visible and ultraviolet portions of the spectrum. The variety of visible energy can be made individually visible to us when sunlight passes through an instrument such as a prism, which refracts the light in differing amounts according to wavelength, thereby separating it into its constituent colours.



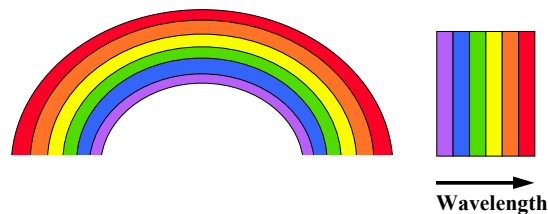
Of the colour that we can see, blue has the shortest wavelength and red the longest. Other colours that we see are combinations of these primary colours.

We see all of these colours because our eyes detect a range of wavelengths, and our brain processes the information into colours. Can you imagine what the world would look like to us if we could only see only very narrow ranges of wavelengths or colour? That is how many sensors work. The information from individual wavelength ranges is gathered, each in a different channel or band. It is up to the image analyst, using a computer, to compile and display the information in a way that is meaningful for us to view. This might be as black and white image, if one range of wavelengths is measured, or as a colour image, if a number of wavelength ranges are measured.

What is a rainbow?



Water droplets act as individual prisms. When sunlight reflects off raindrops, the individual colours in the sunlight are made visible, and distributed according to wavelength because of the refraction of the light. Each wavelength refracts, or bends at a different angle. A rainbow is the result, with shorter wavelengths (violet, blue) in the inner part of the arc, and longer wavelengths (orange, red) along the outer arc.



Colour or label the colours according to wavelength. The next time you see a rainbow, check to see if the order of the colour is the same as described here.

Spectral Response

Sensors detect the reflected energy and with the use of computers and monitors, we are able to display the sensor data as an image. We will explain here how a sensor detects visible and infrared light for example, and how an image is formed from the data. The image can then be manipulated to show a variety of colour combinations.

Absorption (A) occurs when radiation (energy) is absorbed into the target while transmission (T) occurs when radiation passes through a target. Reflection (R) occurs when radiation "bounces" off the target and is redirected. In remote sensing, we are most interested in measuring the radiation reflected from targets. We refer to two types of reflection, which represent the two extreme ends of the way in which energy is reflected from a target: **specular reflection** and **diffuse reflection**. Objects reflect, absorb, and transmit EM (Electromagnetic) radiation. Across the wavelength spectrum, reflections of each target are unique, and can be

characterized by a “signature”. This **spectral signature** makes it feasible for humans and computers to recognize a target, simply by viewing its response to EM energy in a number of different bands.

What is an image?

An image is a visual depiction of the energy reflected from the earth’s surface, as measured by sensors. Images can be produced and analyzed in two formats, analogue (hard copy) and digital (soft copy). Analogue refers to an image captured and stored on film and viewed as a printed-paper copy or film. Digital refers to the fact that the data is composed entirely of numbers, and can be read by a computer. When the numbers are arranged in a specific pattern of columns and lines, they can be represented in an image. (See the Follow-up activity “Raster and Raster Student Activity in the Terrestrial and Marine Ecozones”).

Which type of image format are you using in the MFworks exercise?

Image Information

Each sensor has a particular purpose, whether it is to observe vegetation, general land cover, oceans, or the atmosphere. Certain wavelengths of energy provide the most appropriate information about these applications, and the sensors are specifically designed to measure those wavelength ranges. The detail discernible in an image is dependent on the **spatial resolution** of the sensor and refers to the size of the smallest possible feature that can be detected. **Spectral resolution** describes the ability of a sensor to define fine wavelength intervals. The finer the spectral resolution, the narrower the wavelengths range for a particular channel or band. Finally, the concept of **temporal resolution** is also important to consider in a remote sensing system and it is the frequency of temporal coverage of a sensor/platform system.

The Landsat Thematic Mapper (TM) sensor is a perfect example of such a sensor. Designed as a multipurpose observation satellite, it detects a number of wavelength ranges across the visible and infrared portions of the spectrum. A sensor can detect many ranges of wavelengths at one time. The following describe the wavelengths captured in the Landsat TM bands.

Landsat 5 Thematic Mapper Sensor

Multispectral bands

TM1 (Blue)	0.45-0.52 μm
TM2 (Green)	0.52-0.60 μm
TM3 (Red)	0.63-0.69 μm
TM4 (Near Infrared)	0.76-0.90 μm
TM5 (Shortwave Infrared)	1.55-1.75 μm
TM7 (Shortwave Infrared)	2.08-2.35 μm
TM6 (Thermal Infrared)	10.4-12.4 μm (not used in this study)

Note that each band represents a range of wavelengths, not a single wavelength. The width or range of wavelengths in each band (TM1, TM2, TM3, TM4, TM5, TM6, TM7) determines the **spectral resolution** of a sensor. A sensor measuring a very narrow range of wavelengths such as 0.48 – 0.49 μm for blue wavelengths is an example of fine resolution (range or spectral resolution equals 0.01 μm). Landsat TM Band 1 measures blue wavelengths, but with coarser resolution, a range of 0.45 – 0.52 μm (spectral resolution 0.07 μm).

For each band, the sensor measures the average brightness of **radiance** over each pixel area. This brightness is depicted as one of 256 levels of a grey scale (0 – 255), referred to as the **digital number** of the pixel. Pixels with lower digital numbers are dark; those close to 255 are light. The values of 0 and 255 represent black (0) and white (255) respectively.

The Landsat generation of satellites

The Landsat generation of Remote Sensing satellites has been providing resource applications information since 1972, when this first satellite, referred to then as ERTS-1 (earth resources Technology Satellite) was launched. Designed as an experiment to test the feasibility of collecting multispectral land and coastal information, the successful program has provided numerous datasets used around the world for decades. Landsat's success is due to a combination of useful spectral bands coverage, functional resolution, and wide area coverage. The long lifespan of the program has provided a voluminous archive of earth resource data facilitating long term monitoring and historical records and research. Seven satellites have been launched thus far. The last four satellites carried advanced sensors and have different orbital parameters than Landsats 1,2, and 3. Landsat 7 is currently collecting data. One of the Landsat satellites, Landsat 6 was lost after launch failure so Landsat 5 was still in operation until the launch of Landsat 7 in 1999.

A number of sensors have been aboard the series of Landsat satellites. The most commonly used data are those produced by the MultiSpectral Scanner (MSS) and the Thematic Mapper (TM) sensors. Landsat data is one of the most cost-effective methods of monitoring and analyzing the Earth.

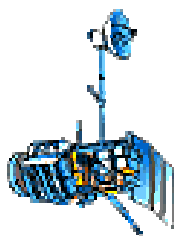
Landsat Application Examples

Just a few of the many applications of Landsat imagery...

- Urban / Land Use planning
- Vegetation and Environmental Monitoring
- Agriculture
- Geology
- Forestry
- Hydrology
- Land cover mapping and monitoring

Landsat data for Canada are received at the Prince Albert and at the Gatineau Satellite Stations. The Canada Centre for Remote Sensing operates these stations.

Landsat 5



Landsat 5 follows a near polar, sun-synchronous orbit 705 km above the Earth. (See on the Canada Centre for Remote Sensing Tutorial the animated explanation of how (<http://www.ccrs.nrcan.gc.ca/ccrs/tekrd/data/introe.html>) satellite data reception works



at the CCRS receiving stations. This tutorial gives an animated representation of the orbit of satellites and explains sun-synchronous). The orbit shifts progressively westward so that every part of the surface of the earth is imaged every 16 days. The extent of coverage is almost global: 81° N to 81° S. Orbital inclination is 99° (near-polar). Sun-synchronization allows the satellite to cross each latitude at the same local time on every orbit. Southbound passes are always made on the illuminated side of the planet; northbound on the dark side. Equatorial crossing southbound is 9:42 am local time. (NASA 1976)

Landsat 5 had two sensors, the Multi-Spectral Scanner (MSS) and the Thematic Mapper (TM).

Landsat Satellite

Altitude of Landsat 5:	705 km
Period:	99 minutes
Inclination:	99°
Number of paths per orbital cycle:	233
Earth coverage:	16 days at the equator

Landsat Sensor used for this exercise

Swath Width (scene width):	185 km (sub-scene 446 x 726 cells)
Spatial Resolution:	30 metres
Radiometric Resolution:	256 grey levels (8 bits)

The TM sensor improved upon the MSS sensor in terms of higher spatial and finer spectral resolution and an increase in number of bands detected. It is the only commercial VIR satellite sensor to detect blue wavelengths. It is a versatile sensor because of the relatively fine spatial resolution, and the spectral coverage.

Landast Spectral Bands Characteristics

Information taken from the Radarsat International web site:

http://www.rsi.ca/classroom/cl_land.htm

Band 1: 0.45-0.52 micrometres (blue-green)

Designed for water body penetration, soil and vegetation discrimination, and forest-type mapping (deciduous/coniferous).

Band 2: 0.52-0.60 micrometres (green)

Matched the green reflectance peak of vegetation. Recommended for vegetation discrimination and plant vigour.

Band 3: 0.63-0.69 micrometres (red)

This band operates in the chlorophyll absorption region and is best for detecting roads, bare soils, and vegetation types.

Band 4: 0.76-0.90 micrometres (near-infrared)

This band is used to estimate biomass. Although it separates water bodies from vegetation and discriminates soil moisture, it is not as effective as TM3 for road identification.

Band 5: 1.55-1.75 micrometres (mid-infrared)

Band 5 is considered to be the best single band overall. It discriminates roads, bare soils, and

water. It also provides a good contrast between different types of vegetation and has excellent atmospheric and haze penetration.

Band 6: 10.5-12.5 micrometres (thermal infrared)

This band responds to thermal (heat) radiation emitted by the target. Thermal radiation is closely related to soil moisture and the height and temperature of vegetation and is best for measuring plant heat stress and thermal mapping. Unlike the other bands, the thermal infrared band has a resolution of 120 m.

Band 7: 2.08-2.35 micrometres (mid-infrared)

This band is useful for discriminating mineral and rock types and for interpreting vegetation cover and soil moisture.

What is a Colour Composite image?

- ❖ *In order to help you do this exercise, refer to the Follow-up Introduction to Remote Sensing the section on Landsat Spectral Bands Characteristics.*

As you have seen, sensors detect and measure a number of bands or ranges of wavelengths across the electromagnetic spectrum. The data resulting from these measurements can form images, which may be viewed in black and white or colour. Each band can be viewed as an individual image in monochrome (one colour – usually grey tones).

The image below shows an individual Landsat Band 1 viewed in grey tones. This image shows information taken in the visible blue area of the electromagnetic spectrum.

Image 1



Figure 1: Band1 (blue light recorder in black and white tones) (0.45 – 0.52 μm)

What are some of the features you can detect on image 1?

The second image is also showing an individual band of the same area of a Landsat image. This band is Band 2 and it shows information taken in the visible green portion of the electromagnetic spectrum.

What are some of the features you can detect easily on this second image?

Compare Band 1 and Band 2. Which one is easier to interpret? Why?

Image 2



Figure 2: Band 2 (green light recorded in black and white tones) (0.52 – 0.60 μm)

The third image is showing an individual band of always the same area of a Landsat image. This band is now Band 3 and it shows information taken in the visible red portion of the electromagnetic spectrum.

What are some of the features you can detect on this third image?

Compare now Band 1, Band 2, and Band 3. Which one is easier to interpret? Why?

Which two bands seem to be similar? Why? Finds some differences between those two Bands?

Image 3



Figure 3: Band 3 (red light recorded in black and white tones) (0.63 – 0.69 μm)

The fourth image is showing again an individual band of the same area of a Landsat image. This band is Band 4 and it shows information taken in the invisible infrared portion of the electromagnetic spectrum.

What are some of the features you can detect on this fourth image?

Explain why the river is black?

Explain why Band 4 is so different then the other 3?

Compare now all Bands (1, 2, 3 and 4). Which one is the easiest to interpret? Why?

Image 4



Figure 4: Band 4 (infrared light recorded in black and white tones) (0.76 – 0.90 μm)

When we want to generate a colour image called colour composite image on computer, we take the data from the three bands, the red band, the green band and the blue. We assign to each band a different colour gun: red, blue and green.

- We are assigning a Blue gun to our Band 1, which is our first image in black and white. You remember that that specific image is representing data from the visible blue portion of the electromagnetic spectrum.
- We are assigning a Green gun to our Band 3, which is our third image in black and white. You remember that that specific image is representing data from the visible red portion of the electromagnetic spectrum.
- Finally, we are assigning a Red gun to our Band 4, which is our fourth image in black and white. You remember that that specific image is representing data from the invisible infrared portion of the electromagnetic spectrum.

COLOUR COMPOSITES

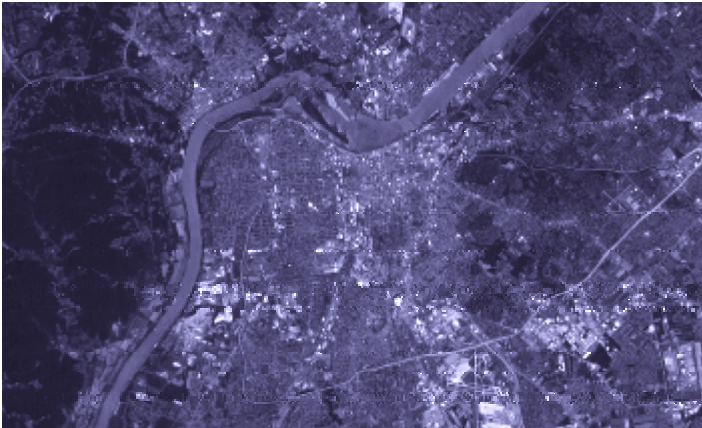


Figure 5: Band 1 displayed in Blue gun



Figure 6: Band 3 displayed in Green gun

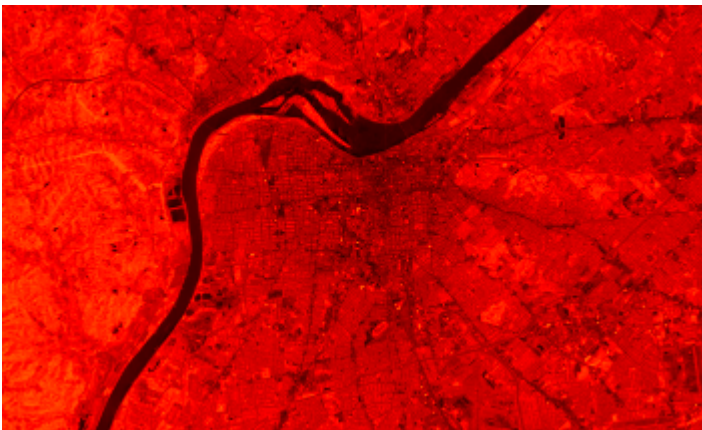


Figure 7: Band 4 displayed in Red gun

The three colour bands are combined into one image to form a colour composite image. In this case, since we had assign the Infrared portion of the electromagnetic spectrum in the Red gun we have produce a colour composite Infrared image.



Figure 8: Colour composite Band 1 blue, Band 3 green, Band 4 red.

Different colour composites can be produced by assigning different bands to the blue, green and red guns.

What type of colour composite image would you produce if you where assigning Band 2 in the Green gun and Band 3 in the Red gun?

Colour composites are also called pseudo or false colour images, because the colours do not necessarily represent the wavelengths being reflected. It is important to remember that the colours are not meaningful, they are just representations of numerical data to aid in visual analysis and interpretation of the remotely sensed information.

Enhancement of an image

There are a number of ways that an image can be made to look more visually appealing. This is important if you wanted to use the image as a base map or poster. The image can also be enhanced to optimize visual interpretation of certain elements, such as urban or industrial areas, land/water boundaries, healthy versus unhealthy vegetation, signs of urban encroachment onto forest or agricultural land, or habitat analysis.

Enhancements are used to facilitate the visual interpretation and understanding of imagery. Although some corrections are made on the image, it may still not be optimized for visual interpretation.

With digital imagery there is the advantage of being able to manipulate the brightness display levels of the image relatively easily. This is accomplished through the use of mathematical model that transform the raw input brightness values of the original image, to the values of brightness in the enhanced image.

There are three main types of enhancements. Contrast Stretches are performed to expand the range of brightness levels used to display the image. Original images usually use only a small portion of the available range of the 256 grey levels. Contrast Stretching involves changing the original values so that more of the grey scale is used, therefore increasing the differences between targets. The other enhancement are Composite Generation which make fullest use of the capabilities of the human eye and Digital Filtering which provides edge enhancement to help remove image blur.

Some characteristics of the Study Area

Study area characteristics

Bedrock: shales in the east, sandstone hills to the south and limestone hills to the west with sinkhole present.

Soils: parent materials are shale, limestone and sandstone. In general the soils are fine loam or loamy clay with a reddish colour. Along the major drainage channels the soils are more clay and have a chocolate-colored topsoil.

Land cover: crops, pasture, hardwood forests (Oak-Hickory association), wood lots, water, towns, roads and airport.

Land use: numerous small family farms; grazing cattle and raising horses; built-up areas; transportation; forestry, and water control.



Land Use and Land Cover of Canada **Follow-up Student Activity Sheets**

Name: _____

****Most of the answers to these questions can be found in the Introduction to Remote Sensing section of the Land Use and Land Cover of Canada Follow-up / Extension.***

1) What other examples of Remote Sensing can you think of?

2) Which type of image format are you using in the MFworks exercise?

3) What are some of the features you can detect on image 1?

4) What are some of the features you can detect easily on this second image?

5) Compare Band 1 and Band 2. Which one is easier to interpret? Why?

6) What are some of the features you can detect on this third image?

7) Compare now Band 1, Band 2 and Band 3. Which one is easier to interpret? Why?

8) Which two bands seem to be similar? Why? Finds some differences between those two Bands?

9) What are some of the features you can detect on this fourth image?

10) Explain why the river is black?

11) Explain why Band 4 is so different then the other 3?

12) Compare now all Bands (1, 2, 3 and 4). Which one is the easiest to interpret? Why?

13) What type of colour composite image would you produce if you where assigning Band 2 in the Green gun and Band 3 in the Red gun?
